Is-Like Presentation

Ortho Team

Bianca, Braxton, Jonathan, Li

Discovery

Clinical Observations

No. Procedures: 9
Total Hours: **46**

No. Procedures: 11
Total Hours: **51**

No. Procedures: 9
Total Hours: **45.5**

No. Procedures: 11
Total Hours: **57**

Total Hours
200
(199.5)

THANKS TO

Dr. Andrew Stein

Dr. Li

Dr. Olszewski

Dr. Freccero

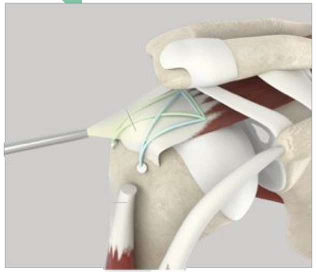
Dr. C Tannoury

Dr. Kain

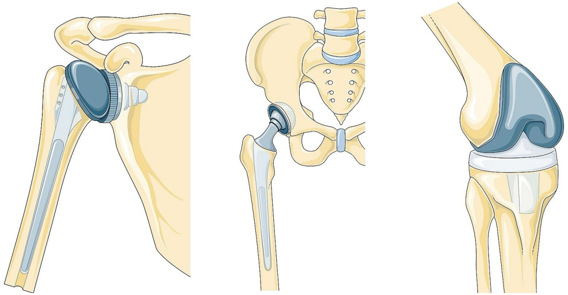
Dr. T Tannoury

And all the **nurses** and **residents** – especially **Dr. Francesca Simeone**, who were all extremely helpful and indispensable to running operations

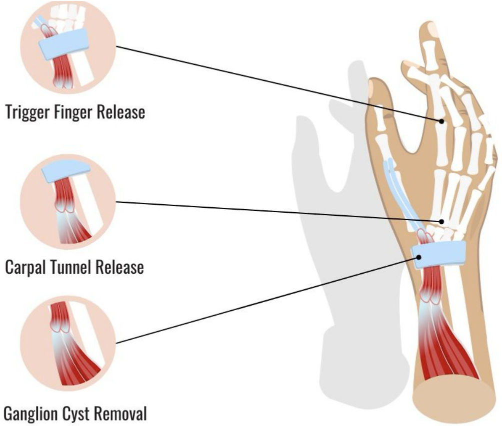
Observed Procedures



Arthroscopic Rotator Cuff Repairs

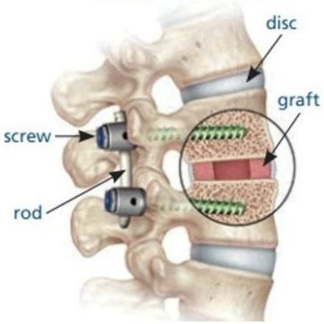


Total and Partial Shoulder, Hip, and Knee Arthroplasties (Joint Replacements)

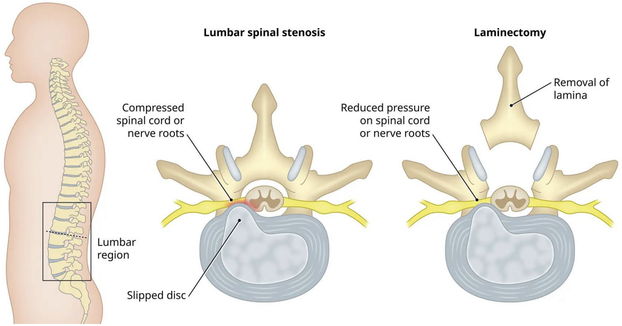


Hand and Wrist Procedures

Internal Fixation and Repairs of Lower Limbs



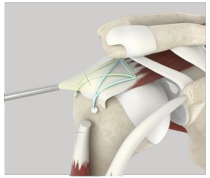
Spinal Fusions



Spinal Laminectomies (Decompressions)

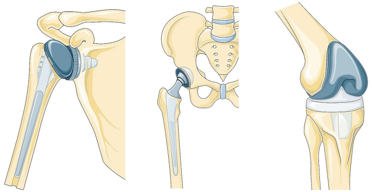
Problem Selection

Rotator cuff retears



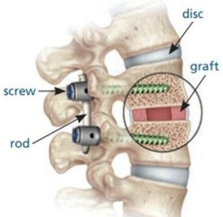
Biological Problem

Joint replacement fitting



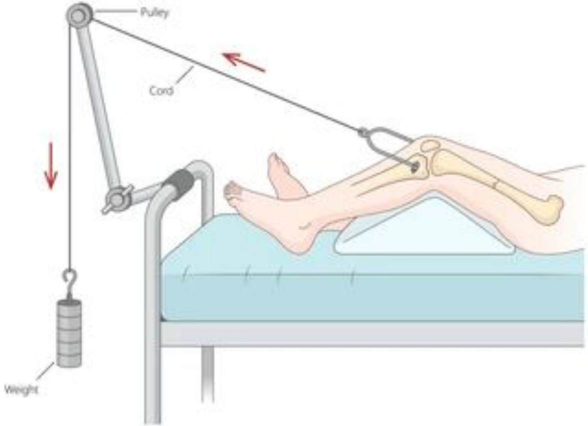
Not a real problem and tight tolerances

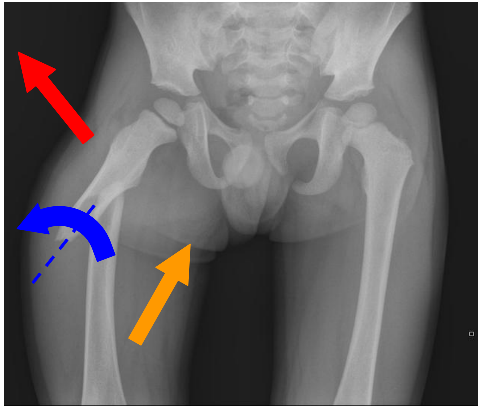
Spinal fusion complications



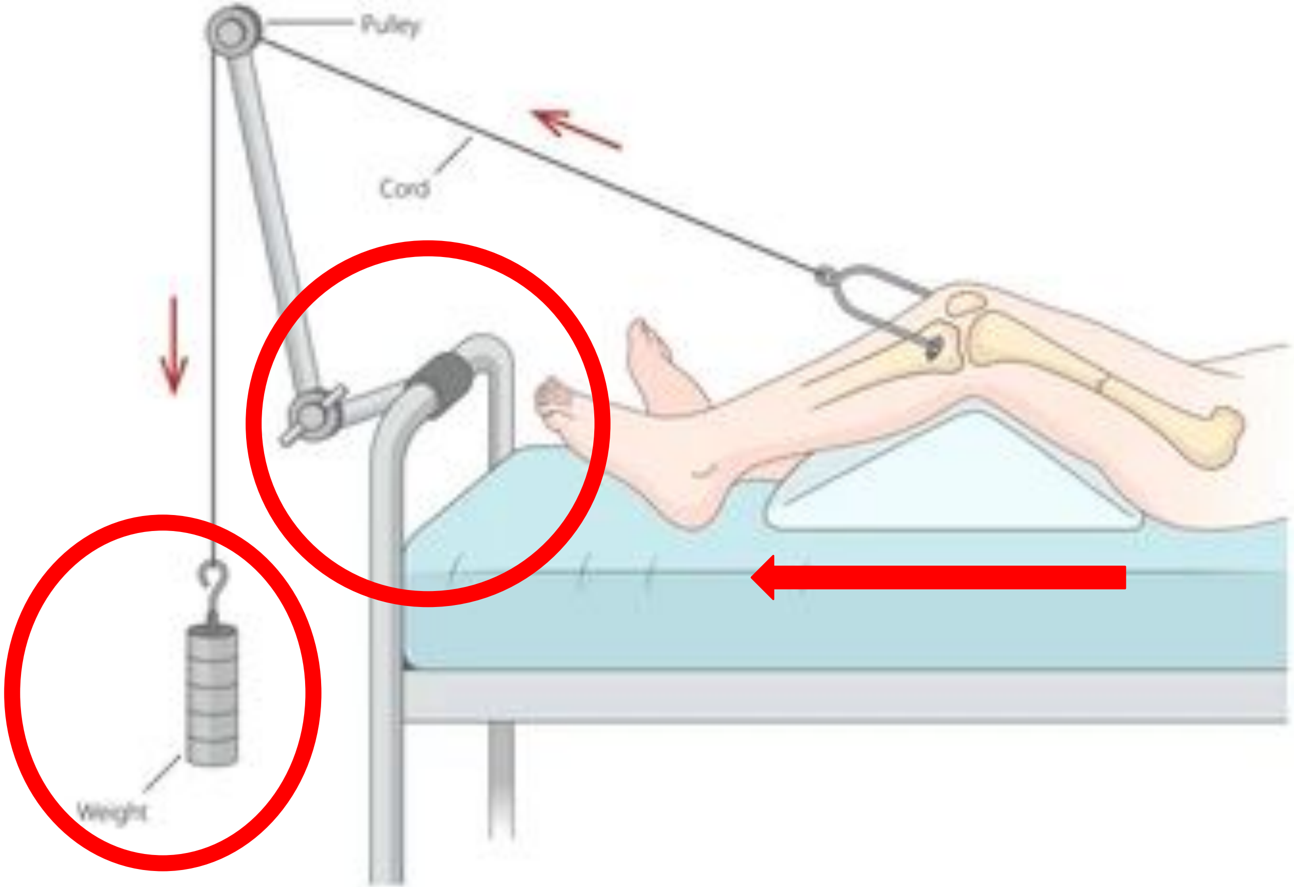
Too complex

Skeletal Traction





Skeletal Traction



Need Statement

Orthopedic surgeons need a way to apply skeletal traction to **adult patients with femoral shaft fractures** that **maintains correct femur length** throughout treatment despite changes in patient position to **prevent femur displacement** that can cause further damage.

Stakeholders

Initiator	Influencer	Gatekeeper	Decider	Purchaser	User	Saboteur
Ortho Surgeons	Nurses/ Residents/ Surgeons	Hospital	Hospital Adm. / Insurance	Hospital Adm./ Insurance	Patient/ Med Staff	Traction Device Companies /Hospitals

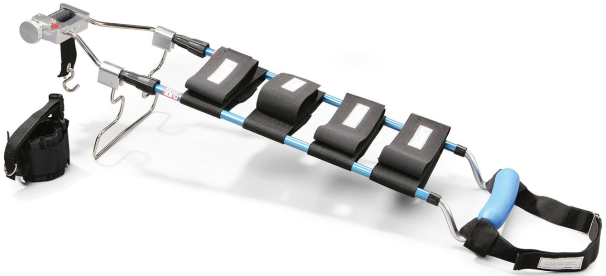
Design

Current Technology Limitations

Expensive!

Either portable OR sufficient traction

No indication of how much force is applied



Proposed Solution

1. No weights
2. Self-contained
3. Adjustable Countertraction

Custom Traction Winch

Cable

Traction Frame

Force Gauge

Leg Cradle and Straps

Countertraction Angle Adjustment Brackets

Value Proposition

Our portable skeletal traction device allows orthopedic surgeons to apply and monitor traction, maintaining correct femur length even **during patient repositioning**.

Unlike current methods, the device is **compatible with most standard hospital beds**, and continued **traction throughout scans** enables more accurate treatment plans for better outcomes.

Development

Device Requirements

	User Need		User Need
1.	Device shall apply traction to femur fracture per current treatment requirements without excessive effort	6.	The pin must align with current traction pin use and safety standards
2.	The device shall maintain the set length and position of femur throughout treatment	7.	Device shall allow for timely evaluation, monitoring, and cleaning of pin site
3.	Device must measure and relay force readings to user	8.	Device pin shall be biocompatible
4.	Device shall be radiolucent for X-rays	9.	Device pin shall be sterilizable
5.	The device must be compatible with use of most hospital beds to allow for efficient patient transport	10.	Device packaging and labeling shall be consistent with FDA standards
		11	Device shall be composed of reusable components

User Need	Design Input
Device shall apply traction to femur fracture per current treatment requirements without excessive effort	Device must be able to apply 5-40 lbs (10-20% of patient's body weight) of traction axially
The device shall maintain the set length and position of femur throughout treatment	Force applied by the device does not deviate more than 2.5 lbs of set force applied
Device must measure and relay force readings to user	Device shall measure applied force with accuracy of ±2 lbs
	Force readings are displayed to user
Device shall be radiolucent for X-rays	Device must be made of components that do not obstruct the fracture during X-rays
The device must be compatible with use of most hospital beds to allow for efficient patient transport	Device fits within bounds of 36 x 80 x 24 inches

Device Inputs and Outputs

Traction Frame -
6061-T6 Aluminum

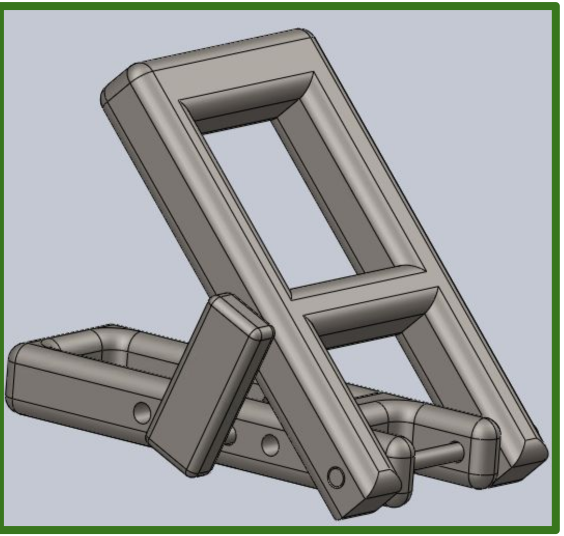
Custom Traction Winch -
Ratchet and pawl w/ worm drive

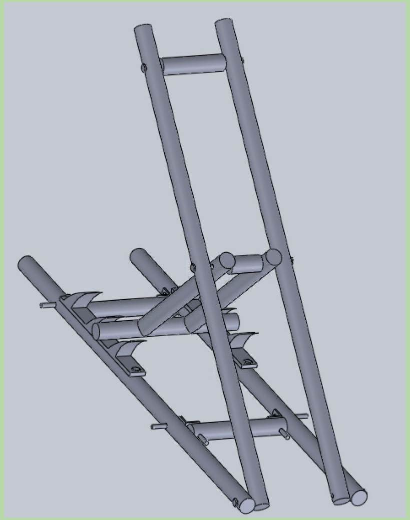
Cable -
Vinyl-coated steel

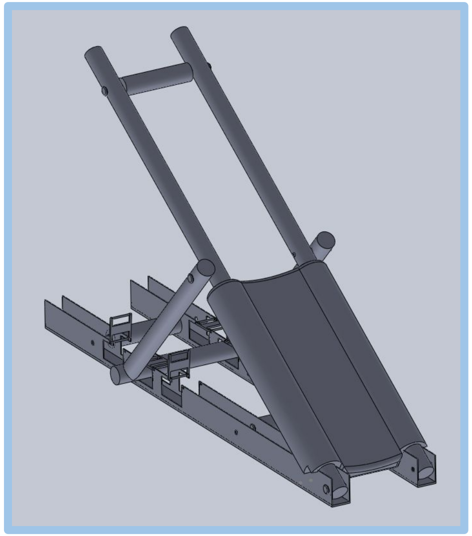
Force Gauge -
Indicates force to user

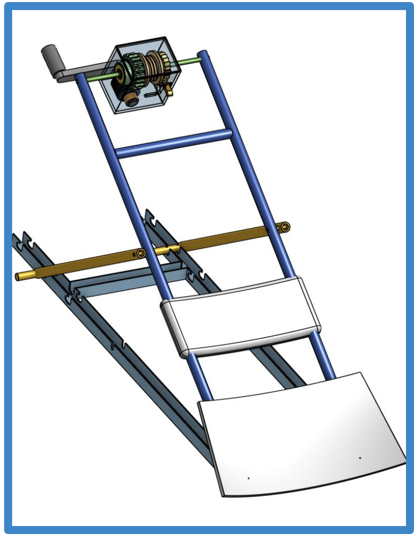
Digital Solidworks Prototyping Iterations

Iteration 1 | Iteration 2 | Iteration 3 | Iteration 4





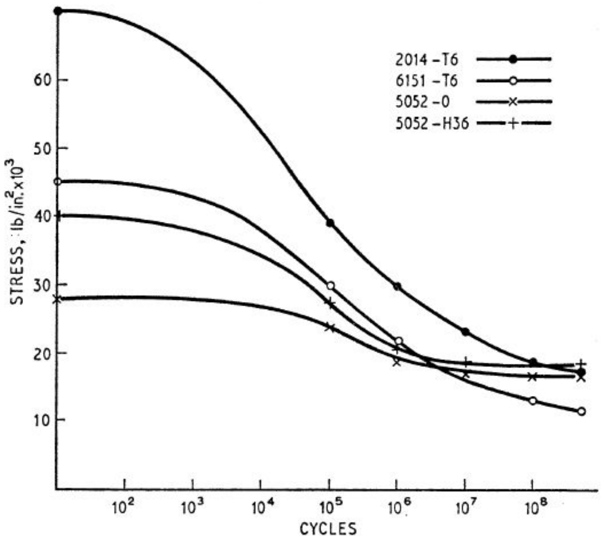




FEA of Validation of Final Model

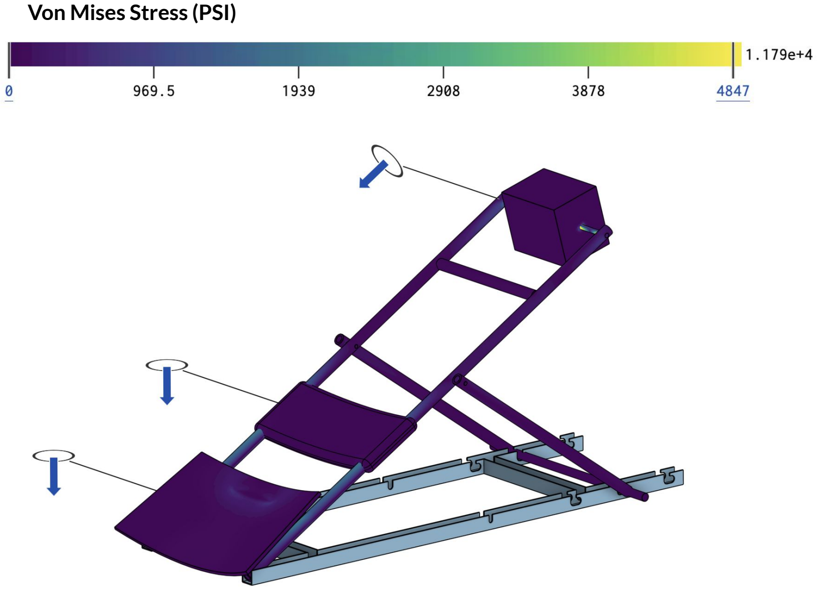
Verification of mechanical properties

1. Aluminum Yield Tolerenace
   a. 40,000 - 70,000 PSI
2. FEA test → stress distribution
3. SN – Curve comparison (Aluminum)



At 4.8 ksi, the design is below fatigue limits (device>10^8 cycles)

Stress Test Analysis



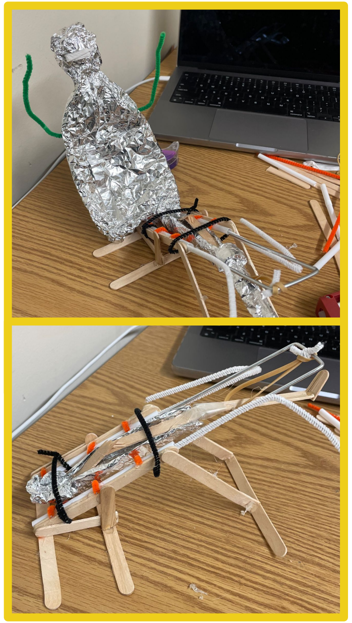
Max Stress of 4847 PSI

Mechanical analysis → device lifespan 5+ years minimum

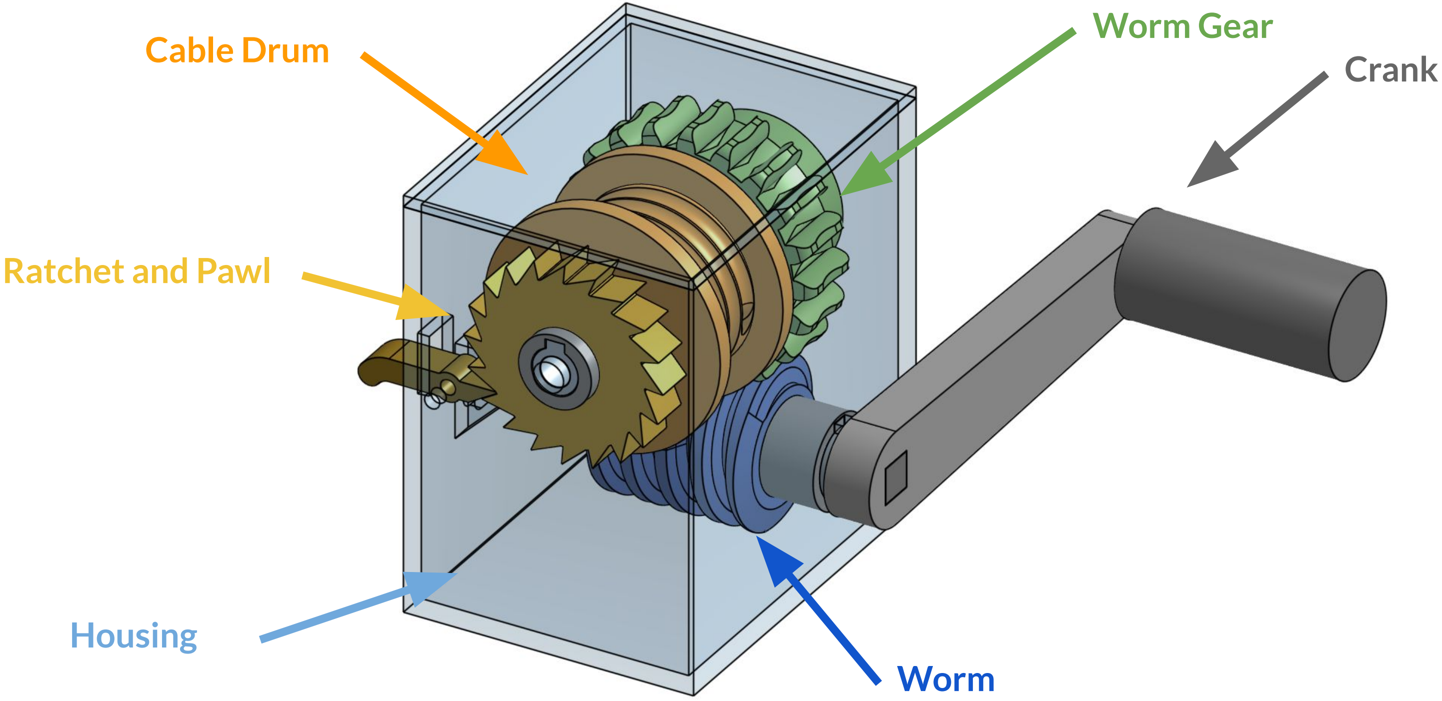
"Aluminum Properties." *Ultrasonic Resonators*, https://www.ultrasonic-resonators.org/materials/aluminum/aluminum_properties.html

Physical Prototyping

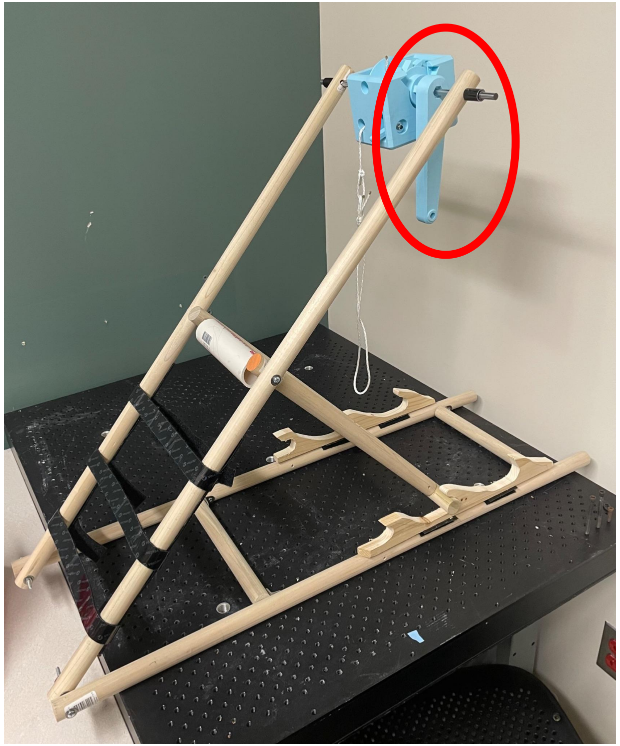
Mockup → **Initial Works Like** → **Advanced Works Like** → **Is Like**

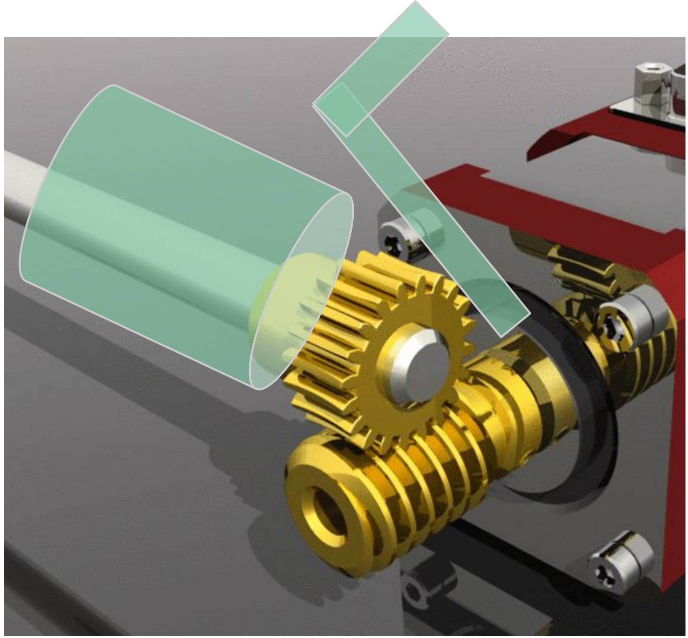


Custom Winch Design



Custom Winch Design

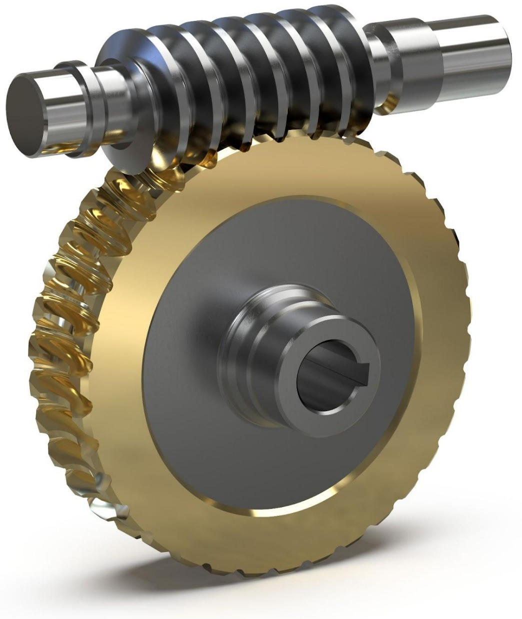




Worm Drives!

Input and output shaft orientation

High torque reduction ratios (used 20:1)



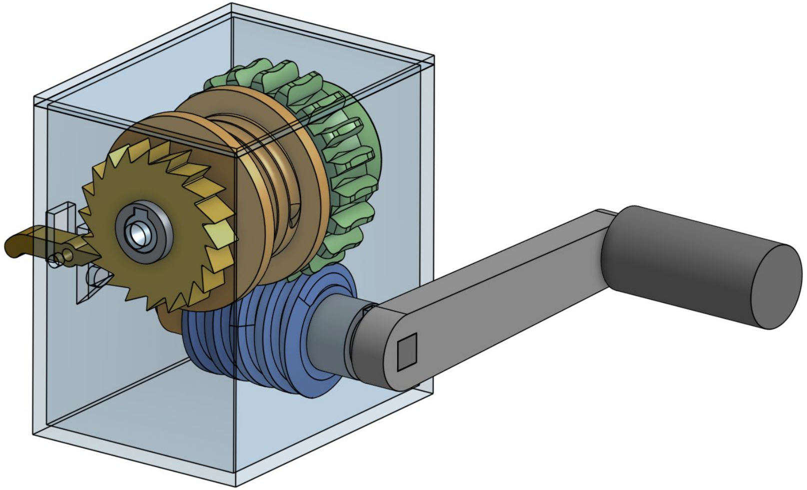
Smooth, incremental adjustments

Self-locking (for lead angles <5°)

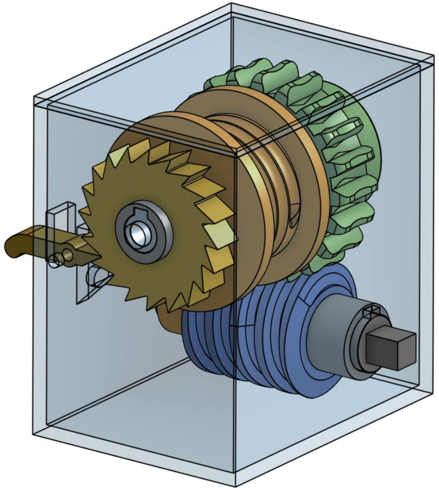
Compact

Risk Analysis and Mitigation

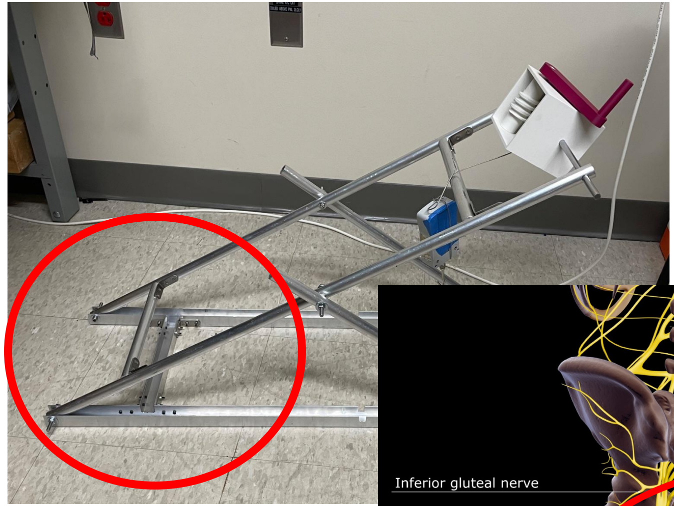
Double Locking with
Ratchet and Pawl

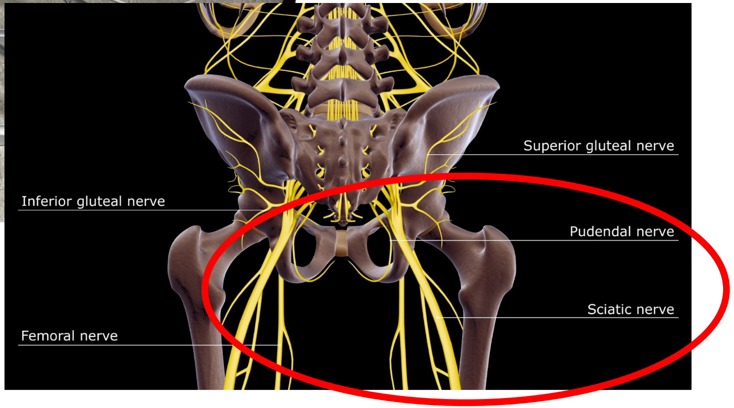


Removable handle to prevent
accidental/untrained adjustment



Risk Analysis and Mitigation



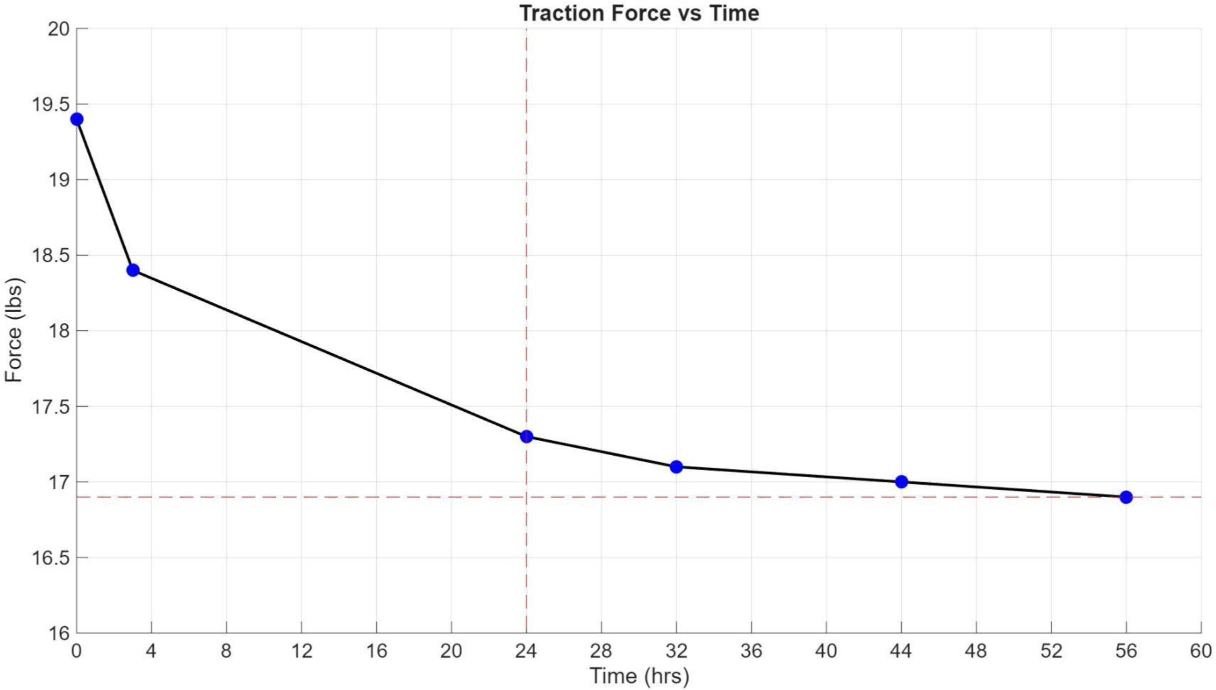


Include leg cradle, padding, and straps

Verification Testing - Maintaining Traction

- 19.4 lb traction force applied at 45 degree setting
- 20 lb leg stand-in attached to force gauge
- Traction force measurements taken in intervals
- Target: ≤ 2.5 lb traction decay over 24 hrs

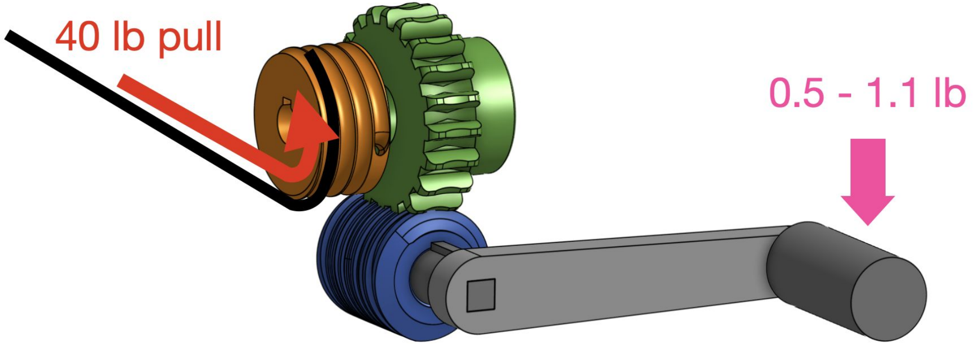
Verification Testing - Maintaining Traction



Verification Testing - Winch Handle

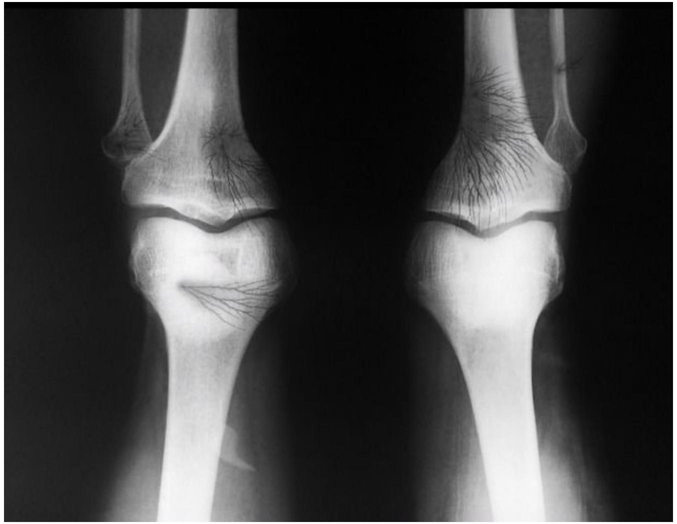
- Measuring force required to turn winch handle using compression force gauge

- Target: < 5 lbs maximum force requirement at 40 lbs of traction



Future Verification - Radiolucency

- Obtain bone equivalent material with small detailed targets
- Image baseline without traction, then material in traction
- Target: Maintain ≥ 80% original SNR, ≥ 3 CNR



Future Verification - Sterility

Frame

- Wipe down test on frame using standard disinfectants
- Target: < 6.4 μg/cm^2 protein residue after cleaning

Pin

- Force testing of pouch opening and cap removal, atmospheric, mechanical, vibrational testing on pouch
- Target: $2 < x < 5$ lb seal on pouch and cap, maintain SAL 10^{-6}

Validation Plan - Traction Workflow

- Assess time needed for staff to apply or take off traction to patient stand-in compared to SOC

- Target: User can set traction at 10, 20, 30, 40 lbs with ± 2 lb accuracy and turn winch with <5 lbs of force

Validation Plan - Frame Portability

- Test whether clinical staff can transfer patient stand-in traction from bed to scanning table

- Target: Staff can complete necessary tasks pertaining to patient transport in <5 min additional time

Validation Plan - X-Ray Compatibility

- Test frame compatibility with standard femur fracture scanning workflow, measuring time needed to complete tasks

- Target: Staff can place pin through bone proxy and can visualize fracture on scans

Deployment

Femur Fractures By The Numbers

US Population in 2021
330 Million

US incidence of femoral shaft fractures
9.5-19 per 100,000

Femoral Shaft Fractures Treated with Traction
~23%*

* varies widely by location

ARBUTUS MEDICAL

COST TABLE

$615 SAVINGS PER CASE

STANDARD WORKFLOW $1,744		WORKFLOW WITH TrakPak® $1,129
$420	Tray Sterilization[1,5] Power System • Battery Pack • Traction Pin • Kirschner Bow	$829 Cost of TrakPak®
$54	Capital Depreciation & Maintenance[2,3,4,5]	
$70	Single Use Materials[5] Prep Materials • Dressing Supplies	
$900	Staff Time Gathering & Opening[5,6,7,8,9]	$100
$300	Staff Time Applying Traction[5]	$200

Market Size

Total Addressable Market (TAM)

10 / 100,000 cases x 330M = 33,000

Revenue/sale	$80+$1800
Sales/year	33,000+330
Total	**$3.23M**

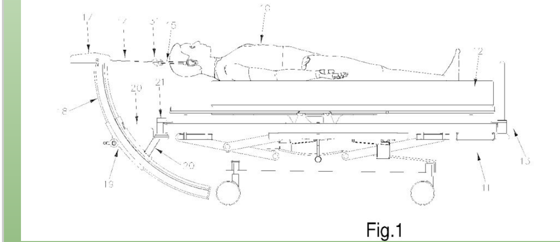
Serviceable Available Market (SAM)

33,000 cases x 40% = 13,200

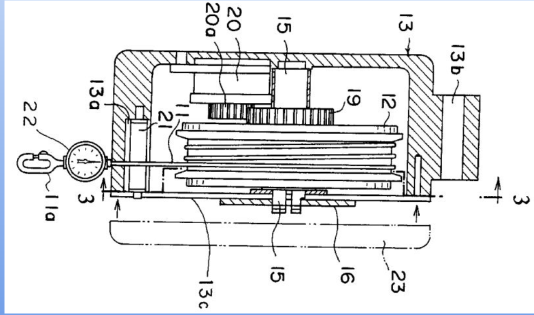
Revenue/sale	$80 +$1800
Sales/year	13,200+132
Total	**$1.29M**

Existing IP Space

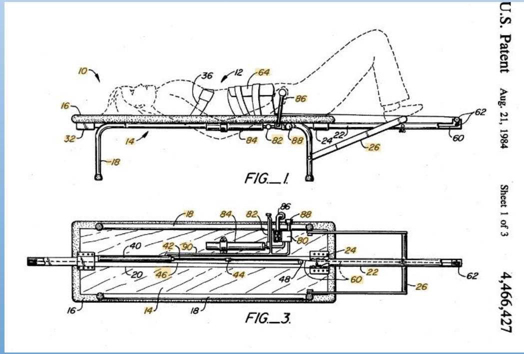
Patent: US6467713B1
Cervical traction with force measurement device
Status: Active



Patent: US6467713B1
Drum-based winch cable traction
Status: Expired



Patent: US3364925A
Cables & pulleys → constant traction
Status: Expired



IP Landscape Overview:

- Thomas Traction → Public Domain (centuries old)
- Improvements → patented
- Expired & fragmented IP across industries → High FTO

Predicate Analysis

Innovation

Tension Application: Free weights → Winch Mechanism

Portability: Fixed bed frame → detached fixture

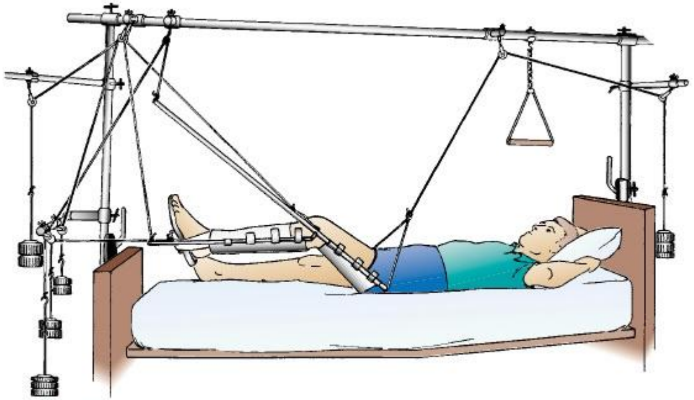
Force Assessment: # Weights → Inline force gauge

Predicate Technology

Thomas Skeletal Traction
(Splint + Steinmann pin)

Novel Technology

Portable Skeletal Traction



Regulatory Pathway

FDA Regulatory Classification

Class 2 →510(k) clearance pathway

External Traction Fixture with Invasive Pin

Substantial Equivalence → Thomas Traction

Relevant Regulatory Standards

ISO 13485 & ISO 14971 – System Quality/Risk

ISO 10993 – Biocompatibility

ASTM F1541 – Skeletal fixation testing

Clinical Testing

Future product differentiation/marketing

De novo pathway → not a target (cost)

Test against Thomas Traction→ Equivalence

Product Codes

Thomas Traction → Grandfathered Device

HSP – Noninvasive traction splints (Fixture)

KTT – Invasive Skeletal fixation (Bone pin)

Business Model Canvas

Execution
- R&D + Mechanical testing
- Small Scale Clinical Trials
- FDA 510(k) clearance

Revenue
- 1st: Device sale
- 2nd: Disposable kits + service
- Optional: Leasing

Customers
- Buyers: Hospitals
- Users: Surgeons, staff
- Stakeholders: Nurses, radiology

Costs
- Upfront: R&D, regulatory
- Ongoing: manufacturing, sales

Go-To-Market
- Direct sales + distributors
- Training + conferences
- Clinical validation

Value
- Portable traction
- Constant force during movement
- Imaging compatibility

Commercialization Strategy

Supply Chain

- Pin designed (in house) → device compatible
- Outsource component manufacturing
- Dual source → supply chain stability

Clinical Benefit Validating Cost

- No weights → reduce failure mode risk
- Imaging compatibility → better surgical outcomes
- Inline force assessment → responsive care
- Portable design → easy mobility across hospital
- Reusable design → cheaper over time for hospital

Unit Cost (COGS)	Device	$803
	Traction Pin	$20
Unit List Price	Device	$1800
	Traction Pin	$80
Gross Margin	Device	55.4%
	Traction Pins	75%

Summary

- Single sale (device) → Reoccuring sale (pins)
- Strong margins → support R&D + growth
- Competitive pricing with multifaceted clinical value
- Positioned for clean market entry and adaptation

Next Steps

Winch Design	Update winch design to add mechanism for limiting maximum allowable force and addressing human factors
Humans Factors Considerations	Improve crank arm-worm shaft interface, and provide large vs small force adjustment options.
Traction Bow and Pin Connection	Consider more details for design of traction bow and pin connection
Testing	Perform imaging and sterility verification, and conduct validation in clinical setting

Additional thanks to:

Our professors

Dr. Mario Cabodi
Professor Megan Lee
Professor Marc Gillette

Our industry mentors

Sam Grossman
Matt Sucec

Questions?